

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB
WEEK 04 TASK: SYSTEM DESIGN DOCUMENTATION
TEAM NAME: CSE4204-8D-T01 || SECTION: 8D

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Section: 8D || **Team ID:** CSE4204-8D-T01

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

1. Team Information & Core Engineering Roles

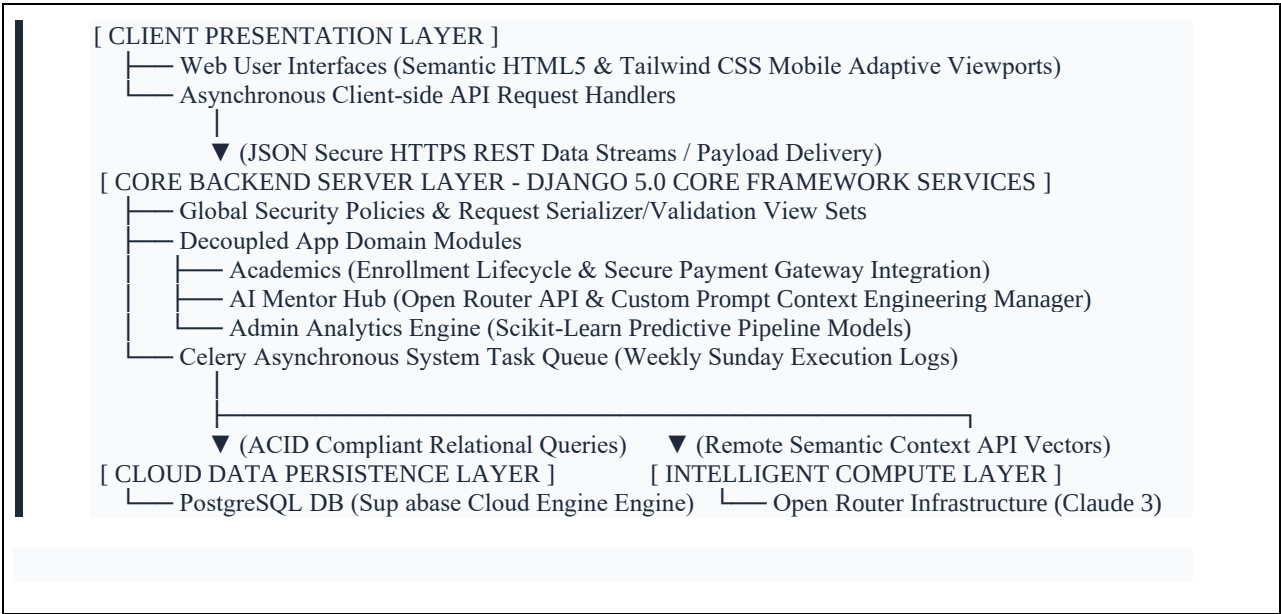
Full Member Name	Student ID	System Design & Engineering Responsibilities
Bashir Alam	11220320963	Team Leader. Core backend system architecture, API design, Celery worker workflow orchestration, Open Router API & ML models integration.
Tisha Rani Chakroborty	11220320954	Frontend Developer. UI/UX layout design systems, interactive dashboard wireframes, and responsive views matching via Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL schema structure planning, entity relationship mapping, and relational table indexing within Supabase.
Ratul Raha Shuvo	11220320949	Backend Support. Payment gateway integrations sub-system design, transactional report templates, and REST API test suite architecture.

2. System Architecture (Optimized Structural Format)

Following industry-standard design paradigms, NUBTK_COPILOT implements a completely decoupled, modular, and asynchronous full-stack architecture. The blueprint guarantees a clean isolation of concerns between front-end presentation templates, Django web controllers, background machine learning analytical processors, and the cloud data layer.

2.1 Visual System Architecture Map

The unified high-level abstract system flow layout is visually structured as follows:



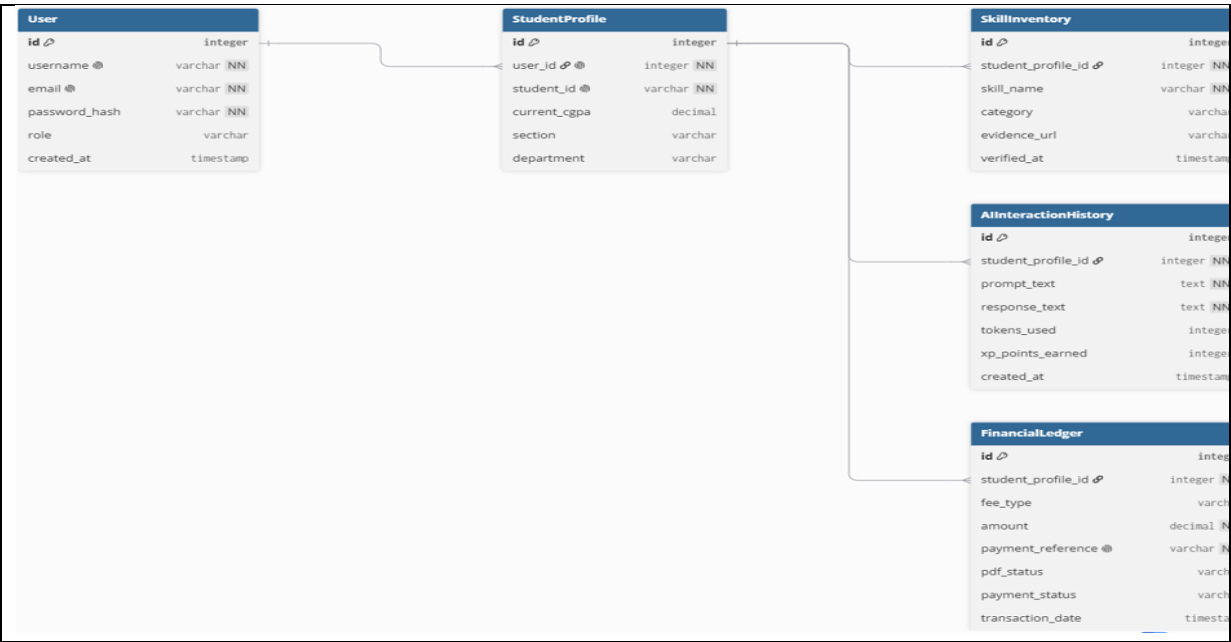
3. Entity Relationship (ER) Diagram Placeholders

The data layer architecture requires strong integrity controls to safely process institutional records, skill items, and transactions. Below is the blueprint of the required relational tables.

3.1 Core Database Schema Schema Blueprint & Definitions

- 1. **User Table (Authentication Core):** Stores account credentials, role types (Student, Faculty, Admin, Super Admin), and encrypted password hashes using secure PBKDF2.
- 2. **Student Profile Table:** Maintains primary academic attributes including explicit unique Student ID registers, Current Cumulative GPA tracking, and assigned Section mapping (Foreign Key to Section).
- 3. **Skill Inventory Table:** Stores student practical technical skill logs (e.g., 'SQL Joins', 'Python Loop Optimization'). Maps to Student Profile (1:M Relationship) and contains verification/evidence links.
- 4. **AI Interaction History Table:** Tracks conversational chat context streams with the AI Mentor Hub, prompt token calculations, and gamified experience point logs.

- 5. **Financial Ledger/Transaction Table:** Manages automated course enrollment registration fees, unique digital payment reference codes, and transactional PDF generation tracking status keys.



User	StudentProfile
id  integer	id  integer
username  varchar NN	user_id   integer NN
email  varchar NN	student_id  varchar NN
password_hash varchar NN	current_cgpa decimal
role varchar	section varchar
created_at timestamp	department varchar

SkillInventory	
id 🔗	integer
student_profile_id 🔗	integer NN
skill_name	varchar NN
category	varchar
evidence_url	varchar
verified_at	timestamp

AllInteractionHistory	
id 🔗	integer
student_profile_id 🔗	integer NN
prompt_text	text NN
response_text	text NN
tokens_used	integer
xp_points_earned	integer
created_at	timestamp

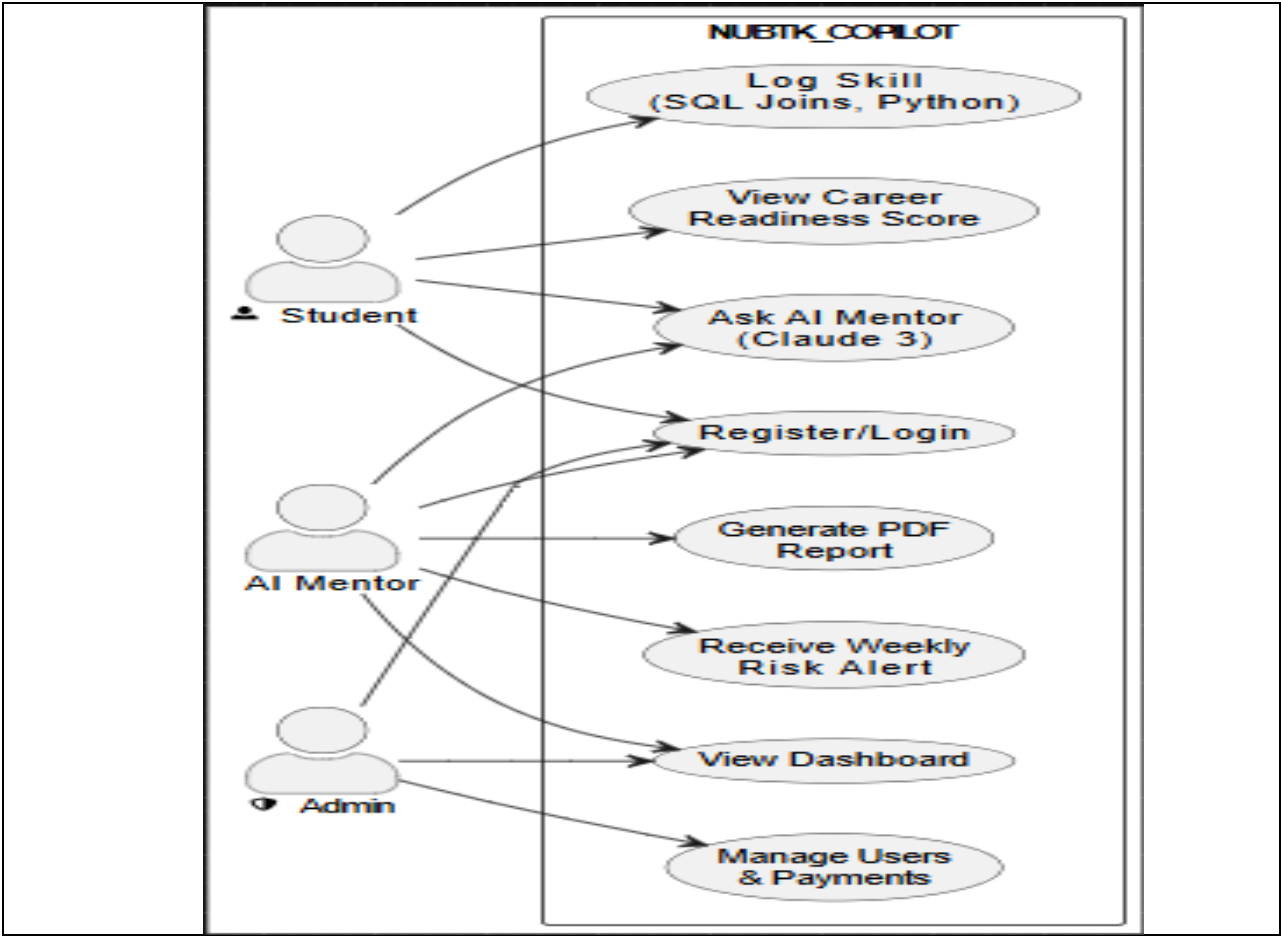
FinancialLedger	
id 🔗	integer
student_profile_id 🔗	integer NN
fee_type	varchar
amount	decimal NN
payment_reference 🔗	varchar NN
pdf_status	varchar
payment_status	varchar
transaction_date	timestamp

4. Use Case Diagram Placeholders

The operational interactions mapping system operations against the three fundamental actor scopes are formalized below. The corresponding design visual box is reserved blank for subsequent structural diagram drafting.

4.1 Functional Actor Scope Interactions Blueprint

- **Student Actions:** Authenticate credentials, log acquired granular computer science skills with evidence verification links, consult the interactive AI Mentor Hub, and review the dynamically evaluated portfolio Career Readiness Score panel.
- **Faculty / Academic Admin Actions:** Access analytics portal views, analyze macro skill matrices across entire classes, review predictive machine learning student danger charts, and receive proactive weekly alert emails.
- **Super Administrator Actions:** Full system config management, institutional authentication global parameters maintenance, security token allocations, and end-to-end automated payment monitoring records.



5. REST API Architecture & Endpoint Specification

To enable seamless full-stack programmatic interfaces, the backend services present a standardized RESTful API framework. All endpoint routing structures enforce standardized payload parameters, data validation formats, and return correct standard HTTP response codes.

HTTP Verb	API Endpoint Routing	Target Scope Module	Expected Input Payload	Primary System Action / Output
POST	/api/auth/register/	Authentication	username, email, password, role	Registers a new system user account inside the institutional DB repository context.
POST	/api/auth/login/	Authentication	email, password	Validates user account password; sets session logs and returns authentication

				token.
POST	/api/skills/log/	Skill Inventory	course_id, skill_name, evidence_url	Appends a verified engineering technical skill node attached to the active student profile.
POST	/api/ai-mentor/chat/	AI Mentor Hub	message_prompt, context_session	Streams context-aware programming guidance payloads generated via OpenRouter Claude 3 models.
GET	/api/analytics/career-score/	Predictive Engine	None (Session Authenticated)	Dynamically calculates and returns the weighted 0-100 index Career Readiness Score array.
POST	/api/finance/pay-fee/	Academics & Payments	student_id, amount, gateway_token	Processes payment transaction records and queues automated background PDF generation jobs.
GET	/api/faculty/dashboard/	Academic Dashboard	None (Faculty Privileged)	Returns aggregated student technical skill inventory trends and machine learning risk graphs.

6. Architectural Deliverables Checklist

- **Decoupled System Architecture Blueprint:** Documented full-stack structural service routing configurations cleanly detailing interaction interfaces.
- **Comprehensive Schema Design Blueprint:** Finalized core relational persistence tables definitions mapping granular skills mapping models.
- **Standardized RESTful Endpoints Mapping:** Structured comprehensive routing matrices outlining correct request payloads validation rules.
- **Multi-Section Submission Formats Adherence:** Ensured proper naming rules conform explicitly to the institutional course guidelines handbook manual.